

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		Indicative content: 1. Wavelength at peak intensity and Wien's Law [can be used to] determine [surface] temperature of the star. 2. Power emitted per square meter or area can be calculated using Stefan's law [details not required] 3. Colour of star can be deduced from wavelength of max intensity / spectrum. 4. Line absorption spectrum shown. 5. Line spectrum arises from passage of [continuous] spectrum/light/radiation / photons through stellar atmosphere 6. absorbing atoms/elements can be identified from wavelength of lines Other relevant points: • Total area under graph represents total power radiated • Absorption spectrum indicates temperature and generation of star • Redshift gives radial / recessional/ velocity/ distance • Reference to inverse square law and distance	6			6		